AROUND THE WORLD IN 19 DAYS

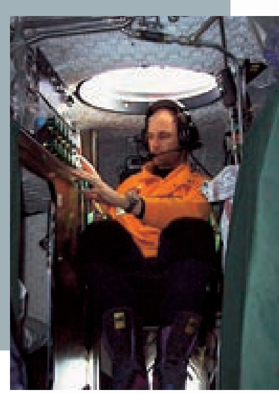
BREITLING ORBITER 3
Orbiter 3 *is the balloon that won the race to fly nonstop around the world. The silver aluminized coating on the envelope acted as insulation, helping keep the gas inside at an even temperature.*

Bertrand Piccard and Brian Jones became the first humans to circle the earth in a balloon on March 20, 1999, as the *Breitling Orbiter 3*, travelling at 130mph (210kph), crossed an invisible finishing line 36,000ft (11,000m) above west Africa.

Orbiter 3 was a hybrid of a helium and hot-air balloon, a type known as a Rozier. The towering silver envelope was filled with helium, while at the bottom a cone-shaped bag was filled with hot air. This was used as variable ballast, allowing the balloonists to gain or lose altitude easily. The gondola was far removed from the wicker basket of early balloonists. It was more like a space capsule, fully pressurized and packed with high-tech equipment.

Like all aspirant around-the-world balloonists of the 1990s, Piccard and Jones took advantage of the jet stream, which from December to March flows eastward in the midlatitudes at speeds of up to 200mph (320kph). Starting from Chateau d'Oex, Switzerland, they flew south across Spain to pick up the jet stream over north Africa. From then on, manipulating the hot air to make the balloon rise and fall, they were able skillfully to locate winds that carried them on a remarkably direct route east, at around 20 degrees north of the equator. They spent much of the journey at above 30,000ft (9,100m), enduring very low temperatures.

Hoping to finish the voyage in style, Piccard and Jones tried to reach the Pyramids at Giza, but ended up coming down in the Sahara and waiting hours for a helicopter to pick them up. Their epic journey had taken 19 days, 21 hours, and 55 minutes.



INSIDE VIEW
Inside the gondola of the Breitling Orbiter 3 *most of the available space was devoted to highly advanced communications and navigation equipment.*

GLOBAL CHALLENGERS
Left to right, Steve Fossett, Richard Branson, and Per Lindstrand pose in front of the balloon ICO Global Challenger, *in which they failed to fly around the world in 1998. Both Fossett and Branson were rich men who repeatedly risked their lives in pursuit of the around-the-world record.*

Solar Challenger flew 163 miles (261km) from just outside Paris to Manston, Kent, in the south of England. Battery-assisted manpower became an important area of experimentation, with aircraft achieving speeds in excess of 30mph (50kph). But human-powered flight remained the preserve of the extremely fit. It was also sensitive to weather conditions – any strong wind would destroy the fragile aircraft in no time.

Record-breaking ballooning

Another activity that attracted adventurers and experimenters in the late 20th century was ballooning. Although interest in the first form of manned flight had never totally faded, it had long appeared a technology incapable of much further development. However, in 1931 Swiss physicist Auguste Piccard invented a pressurized gondola that allowed balloonists to ascend to previously fatal heights. Balloon ascents became both a scientific means of exploring the upper reaches of the atmosphere and a field of international competition to establish altitude records.

In 1935, the American *Explorer II* balloon, crewed by Albert Stevens and Orville Anderson, reached an altitude of 72,395ft (22,056m), a record that was not broken for two decades.

More exciting from a media and sporting point of view was long-distance ballooning. All the spectacular record-breaking achievements of the airplane pilots of the 1920s and 1930s – first transatlantic flight, first nonstop transcontinental flight, and so on – stood to be repeated in unpowered lighter-than-air craft. By the 1970s, balloon technology and, crucially, the understanding of winds and weather patterns, had